*Time Series Analysis & Forecasting*

*MPBA G512*

# *Adaptive GARCH-Based Volatility Modeling and Monte Carlo Risk Estimation for USD/INR*

Project Report

Submitted To

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# TABLE OF CONTENTS


| Contents | Page No. |
|---|---|
| Introduction, Problem Statement, Objectives and Literature Review | 2 |
| Methodology and Model Specification | 4 |
| Data Description | 9 |
| Hypothesis Testing and Results | 11 |
| Findings and Interpretation | 21 |
| Conclusion | 26 |
| References | 28 |
| Appendix (Code & Dataset) | 29 |

# Introduction, Problem Statement, Objectives and Literature Review

## 1.1 Introduction

Financial time series, especially exchange rate returns, show well-known patterns like volatility clustering, conditional heteroskedasticity, and excess kurtosis. These traits go against the basic ideas of constant variance and normality that classical time series models are based on. This means that traditional linear methods can't be used to model financial risk. Volatility in these series changes over time and needs to be modeled as a random process.

In foreign exchange markets, these changes are caused by a mix of big-picture economic factors, capital flows, speculation, and government actions. Because of this, exchange rates are not only unstable, but they may also react unevenly to shocks in the market. This is especially true for currencies in emerging markets, like the Indian Rupee (INR), where structural factors and central bank actions can change the size and direction of volatility responses.

The USD/INR exchange rate is very important for India's economy as a whole. It affects trade competitiveness, inflation, and the balance sheets of businesses. Companies that deal with foreign currencies have a lot of uncertainty in the short term, so it's important to accurately model exchange rate risk for hedging and financial planning.

However, standard risk estimation approaches based on constant variance assumptions fail to capture the dynamic and potentially asymmetric nature of volatility in such markets.

In contrast to conventional approaches that directly impose volatility models, this project adopts a comprehensive pre-diagnostic framework prior to model estimation. Specifically, the analysis incorporates formal statistical testing for stationarity (Augmented Dickey-Fuller test), normality (Jarque-Bera test), serial correlation (Ljung-Box test), and volatility clustering (ARCH-LM test). This ensures that the chosen model specification is fully supported by the empirical characteristics of the data rather than being imposed a priori.

## 1.2 Problem Statement

The traditional methods for modeling risks have considered the assumption of independent and identically distributed financial returns with constant volatility and symmetry of the effect of shocks. Financial market evidence proves otherwise, as it shows that the volatility of returns is time-dependent and clustering, and the effect of shocks can be asymmetric depending on the nature of the shocks.

Failure to consider these features while modeling the USD/INR exchange rate may cause an underestimation of tail risks during turbulent periods due to the presence of heteroskedasticity and asymmetry in financial markets.

This research examines the modeling of time-varying and asymmetric USD/INR volatility as well as the effects thereof on the assessment of multi-period risk metrics.

## 1.3 Objectives

The primary objective of this study is to model the conditional volatility of USD/INR exchange rate returns using GARCH-class models.

The specific objectives are:

- To estimate volatility dynamics using the GARCH(1,1) framework
- To test for the presence of **asymmetric volatility effects** using the GJR-GARCH model
- To implement a **data-driven model selection approach** based on statistical significance
- To account for **fat-tailed return distributions** using the Student-t specification
- To validate model assumptions through a structured pre-diagnostic testing framework
- To generate forward-looking risk estimates using **Monte Carlo simulation**
- To compute and analyze **Value at Risk** (**VaR**) and **Expected Shortfall** (**ES**) over a 30-day horizon

## 1.4 Literature Review

The concept of modeling of volatility in time was introduced by Bollerslev (1986) with the help of GARCH models. GARCH allows modeling of conditional variance as a function of past squared residuals and conditional variance.

Nevertheless, the traditional approach assumes that shocks do not affect volatility asymmetrically, while GJR-GARCH introduced by Glosten, Jagannathan, and Runkle (1993) takes account of the fact that shocks impact the financial markets asymmetrically. In particular, negative shocks will cause different reaction in volatility compared to that of positive shocks.

In financial markets, empirical data indicates that the probability of occurrence of extreme cases in the form of fat-tails is greater compared to that estimated by the normal distribution. Hence, heavy-tailed distributions such as Student-t distribution play significant role in volatility estimation and modeling. Monte Carlo simulation provides a flexible approach for estimating these risk measures by generating the full distribution of future returns under time-varying volatility conditions.

Recent empirical approaches emphasize the importance of diagnostic validation prior to model estimation to ensure statistical adequacy, particularly in financial time series exhibiting non-normality and volatility clustering.

## 2. Methodology and Model Specification

### 2.1 Pre-Diagnostic Testing Framework

Before estimating the volatility model, a comprehensive diagnostic framework is applied to validate the statistical properties of the data.

The following tests are conducted:

• Stationarity: Augmented Dickey-Fuller (ADF) test

- Normality: Jarque-Bera test
- Serial Correlation: Ljung-Box test on returns
- Volatility Clustering: Ljung-Box test on squared returns and ARCH-LM test

This framework ensures that the modeling approach is not imposed arbitrarily, but is instead supported by empirical evidence from the data. Only after confirming these properties is the GARCH modeling framework applied.

## 2.1 Return Specification

Let $P_t$ denote the USD/INR exchange rate at time $t$. The return series is defined as the continuously compounded log return:

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Log returns are preferred over simple returns due to their time-additive property and approximate stationarity. For high-frequency financial data such as daily exchange rates, the return series typically exhibits negligible autocorrelation in the mean but significant dependence in higher moments, particularly variance.

## 2.2 Conditional Mean Equation

The return process is modeled as:

$$r_t = \mu + \epsilon_t$$

where $\mu$ denotes the unconditional mean of returns and $\epsilon_t$ represents the innovation term. Considering the weak dependence noted between daily exchange rates returns, there will be no inclusion of an autoregressive term in the mean equation. As such, this analysis will mainly focus on modeling the conditional variance of the innovation term $\epsilon_t$.

Even though the Ljung-Box test suggests a slight presence of autocorrelation in the return series, the degree of correlation is practically inconsequential to the extent of volatility estimation. As such, an ARMA(0,0) specification will be adopted.

The necessary condition for the covariance stationarity of the GJR-GARCH specification requires that the variance series satisfies the constraint $\alpha_1 + \beta_1 + 0.5\gamma_1 < 1$. This constraint will ensure convergence of volatility shocks, hence avoiding the possibility of having explosive variance forecasts.

## 2.3 GARCH(1,1) Model

The conditional variance of returns is modeled using the GARCH(1,1) specification:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where:

$\omega > 0$ represents the long-run variance level

$\alpha \geq 0$ captures the impact of recent shocks (ARCH effect)

$\beta \geq 0$ captures volatility persistence (GARCH effect)

The GARCH model takes into account volatility clustering, implying that after big shocks, there is a period of high volatility. Volatility persistence can be defined using $\alpha+\beta$ coefficients, and high values of $\alpha+\beta$ imply that shocks have a prolonged effect.

ARCH effects are confirmed statistically before applying the model.

## 2.4 GJR-GARCH Model (Asymmetric Extension)

To capture potential asymmetry in volatility response, the GJR-GARCH(1,1) model is employed:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \gamma\epsilon_{t-1}^2 I_{t-1} + \beta\sigma_{t-1}^2$$

where the indicator function $I_{t-1}$ is defined as:

$$I_{t-1} = \begin{cases} 1, & \text{if } \epsilon_{t-1} < 0 \\ 0, & \text{otherwise} \end{cases}$$

The parameter $\gamma$ captures the asymmetric response of volatility to shocks. A statistically significant $\gamma$ indicates that the magnitude of volatility depends on the sign of past innovations, thereby extending the symmetric GARCH framework.

## 2.5 Hypothesis Testing for Asymmetry

The presence of asymmetric volatility is formally tested using the coefficient $\gamma$ in the GJR-GARCH model:

$$H_0: \gamma = 0 \text{ (no asymmetry)}$$
$$H_1: \gamma \neq 0 \text{ (asymmetry present)}$$

The decision rule is based on the p-value of the estimated coefficient. If the null hypothesis is rejected at the 5% significance level, the GJR-GARCH specification is retained. Otherwise, the model simplifies to the standard GARCH(1,1), ensuring parsimony.

This approach ensures that model selection is data-driven and statistically justified, rather than imposed *ex ante*.

## 2.6 Distributional Assumption

The innovation term is modeled as:

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu(0,1)$$

where $z_t$ follows a standardized Student-t distribution with $\nu$ degrees of freedom. This specification allows the model to capture fat-tailed behavior observed in financial returns, where extreme events occur more frequently than predicted by the normal distribution.

The parameter $\nu$ controls the thickness of the tails, with lower values indicating heavier tails.

This specification is empirically justified by the Jarque-Bera test, which strongly rejects normality and confirms the presence of fat tails in the return distribution.

## 2.7 Model Selection Framework

The modeling approach follows a two-stage framework:

**Diagnostic Stage:**
Estimate a GJR-GARCH(1,1) model to test for the presence of asymmetry.

**Selection Stage:**
If $\gamma$ is statistically significant → retain GJR-GARCH
If $\gamma$ is not significant → revert to standard GARCH

This framework ensures that the final model balances statistical adequacy and parsimony. Information criteria such as AIC and BIC are used as secondary validation measures.

The model selection process is supported by diagnostic testing, ensuring that GARCH modeling is applied only when statistically justified by the presence of ARCH effects.

## 2.8 Monte Carlo Simulation

To estimate multi-period risk, a Monte Carlo simulation approach is employed using the fitted GARCH model. The simulation generates $N = 10{,}000$ return paths over a horizon of $h = 30$ days.

Each simulated path is generated recursively using the estimated conditional variance dynamics. The cumulative return over the horizon is defined as:

$$R_{t,h} = \sum_{i=1}^{h} r_{t+i}$$

The simulation is initialized using the last observed volatility state, ensuring that forecasts reflect current market conditions rather than long-run averages.

## 2.9 Risk Measures

Risk is quantified using two standard measures:

**Value at Risk** (**VaR**)

$$VaR_a = Quantile_a(R_{t,h})$$

VaR represents the maximum expected loss over a given horizon at confidence level $\alpha$.

**Expected Shortfall** (**ES**)

$$ES_a = E[R_{t,h} \mid R_{t,h} < VaR_a]$$

Expected Shortfall measures the average loss in the tail beyond the VaR threshold, providing a more comprehensive assessment of extreme risk.

# Data Description

## 3.1 Data Source and Sample

[Dataset Link]

Exchange rate series data were collected in terms of the daily USD/INR exchange rates obtained from Federal Reserve Economic Data with the ID code of DEXINUS representing the number of Indian Rupee per US dollar.

A total of 2,489 exchange rate data was obtained for a duration of about 10 years running from April 2016 to April 2026. Exchange rate data does not exist on the weekends and some holidays; therefore, the obtained dataset only contains trading days in the sample period.

The data used in this research consist of a total of 2,490 exchange rate series data within the period between 2016 and 2026.

## 3.2 Data Transformation

Transformation is done to turn the non-stationary exchange rate data into continuously compounded log return series:

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Such transformation results in an approximately stationary series ready for GARCH modeling and log returns series adds up across periods; therefore, it allows the series to be applied in Monte Carlo simulation. Missing data were deleted using listwise deletion.

Exchange rate series data is a non-stationary process which requires a transformation to make it suitable for time series modeling. Log returns computation makes it more stable with regard to mean and variance.

## 3.3 Statistical Properties

This return series displays important features of financial time series data. The mean of the returns series is near zero, reflecting the weakly predictable nature of movements in the foreign exchange market. Nevertheless, the variance of returns is time varying, revealing conditional heteroscedasticity in the process.

In addition, one can expect excess kurtosis in the distribution of returns since this means that the distribution of returns has fatter tails than those of a normal distribution. Hence, there is the justification for using the Student-t distribution in the conditional variance model.

This empirical feature, among others, explains the necessity of modeling returns with a model belonging to the class of GARCH models. Formal statistical tests reveal that the return series is stationary, is non-normal, and displays strong clustering of volatility, making the choice of GARCH models appropriate.

| Parameter | Value |
|---|---|
| Data Source | Federal Reserve Economic Data (FRED) — DEXINUS |
| Quotation Convention | INR per USD (direct quote from Indian perspective) |
| Sample Period | 15 April 2016 – 3 April 2026 |
| Total Observations | 2,489 daily log-return observations |
| Return Transformation | Log-differenced: $r_t = \ln(P_t / P_{t-1})$ |
| Current Spot Rate | ₹93.09 per USD (as of April 2026) |
| 10-Year Range | ₹64.84 (Apr 2016) to ₹87.67 (high in 2025) |
| Mean Daily Log Return | ~0.000045 (near-zero, consistent with random walk) |
| Annualised Volatility (approx.) | ~4.5–5.0% — compressed by RBI intervention |

# 4. Hypothesis Testing and Results

## 4.0 Pre-Diagnostic Results

Prior to estimating the volatility model, the stationarity of the returns series is tested through the Augmented Dickey-Fuller (ADF) Test. The null hypothesis of the ADF test is that there exists a unit root in the series, which can be stated as:

$H_0$: There exists a unit root in the series (series is non-stationary).

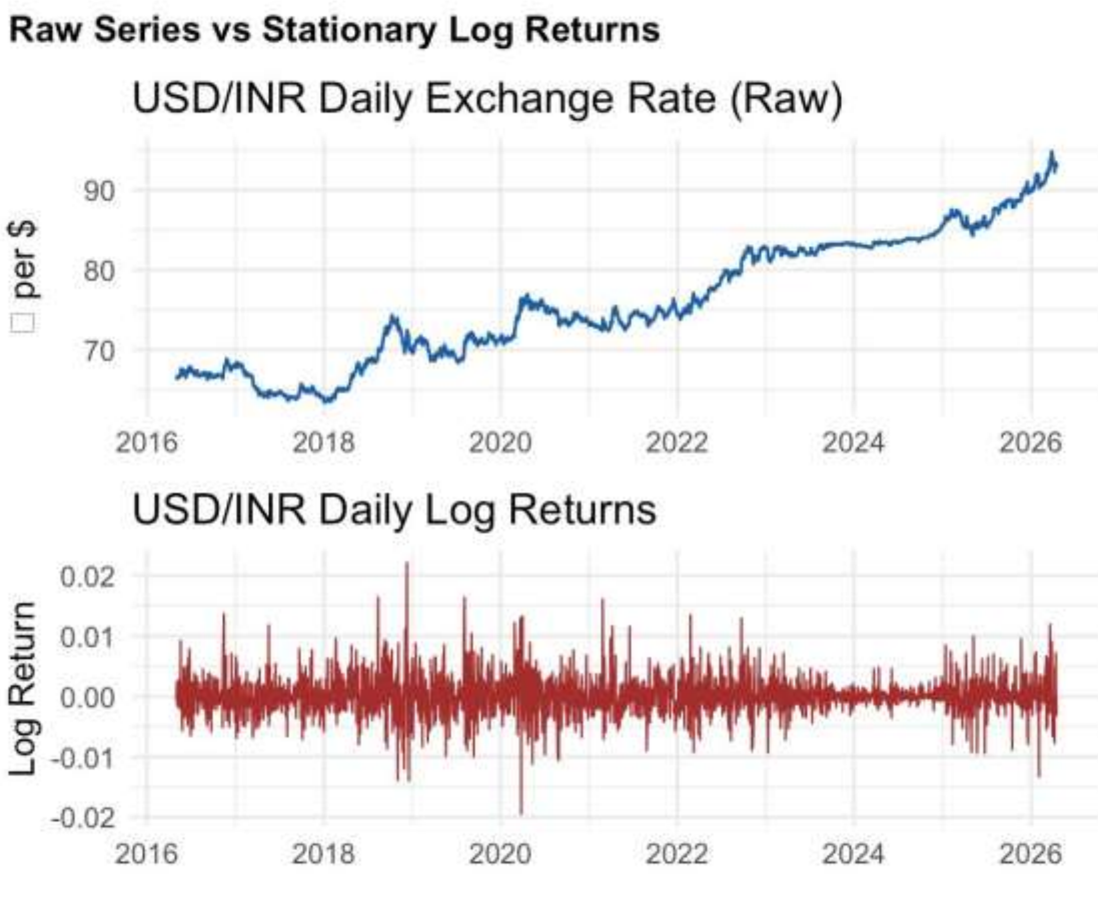



Figure 1: Raw USD/INR Exchange Rate vs Stationary Log Returns

### 4.1.1 Stationarity (Augmented Dickey-Fuller Test)

**$H_0$:** The series contains a unit root (non-stationary)

**$H_1$:** The series is stationary

The ADF test on the raw USD/INR price series yields a statistic of −2.77 ($p = 0.254$), confirming non-stationarity. After log-differencing, the ADF statistic drops to −13.02 ($p < 0.01$), firmly rejecting the null. Log returns are thus confirmed stationary and suitable for GARCH estimation.

### 4.1.2 Normality (Jarque Bera Test)

**$H_0$:** Returns are normally distributed

**$H_1$:** Returns are non-normal (fat tails / skewness)

The Jarque-Bera statistic of 1,954.76 ($p < 2.2 \times 10^{-16}$) overwhelmingly rejects normality. With excess kurtosis of 7.29, the distribution is heavily leptokurtic. This result mandates the use of Student-t distributed innovations in the GARCH model rather than Gaussian errors.

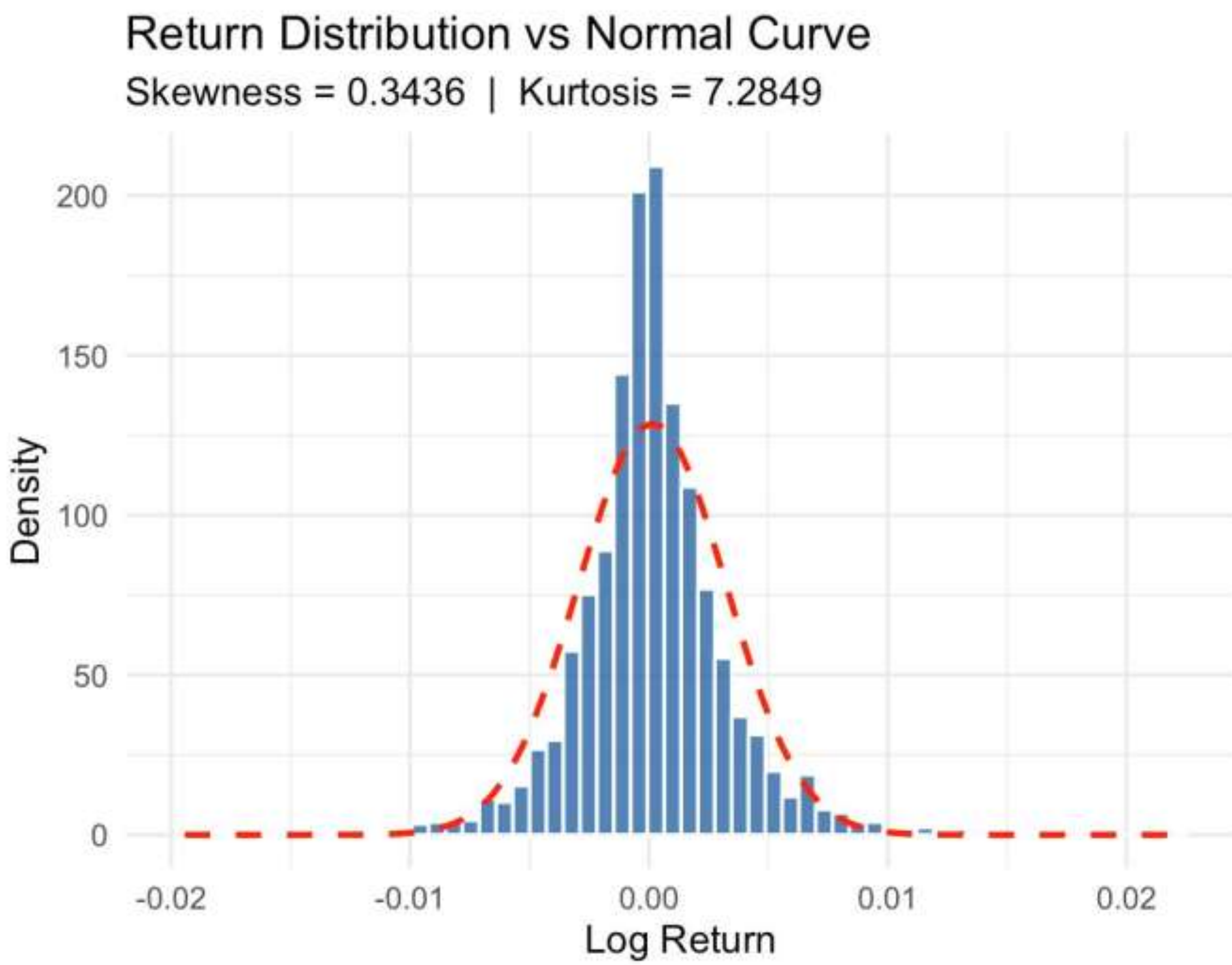



Figure 2: Distribution of Log Returns with Heavy Tails

### 4.1.3 Serial Correlation and Mean Model Selection (ARMA(0,0))

$\mathbf{H_0}$: No autocorrelation in log returns - ARMA(0,0) mean model is sufficient

$\mathbf{H_1}$: Autocorrelation exists in returns - ARMA terms required in the mean equation

The Ljung-Box Q-statistic on log returns at lag 12 equals 24.95 ($p = 0.015$), which rejects the null at the 5% level. This is a statistically significant result that warrants a deliberate mean model selection decision rather than automatic addition of ARMA parameters.

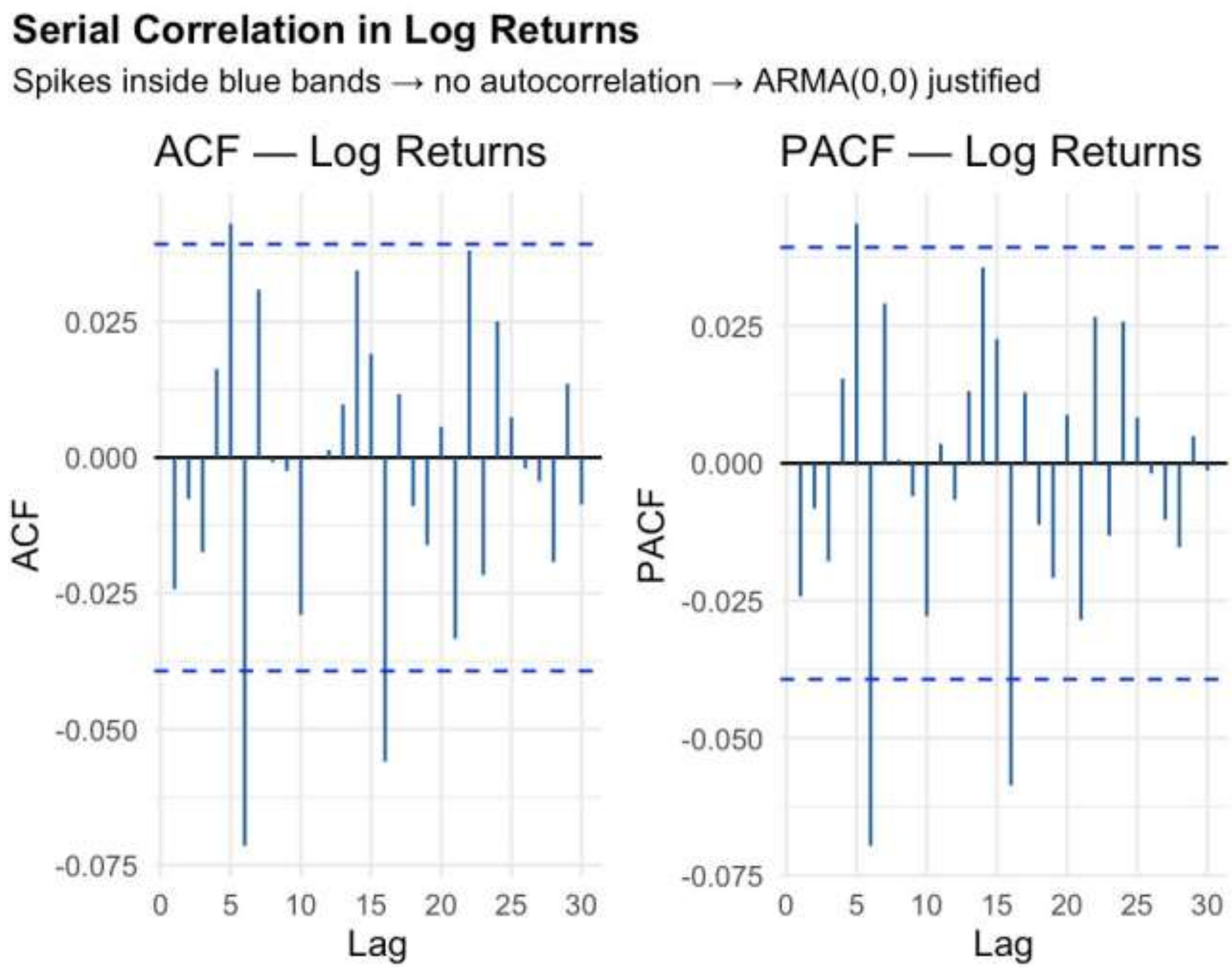



Figure 3: ACF and PACF of Log Returns

### 4.1.4 ARCH Effects (Ljung-Box on Squared Returns + ARCH-LM)

**$H_0$:** No ARCH effects (homoskedastic, constant variance)

**$H_1$:** ARCH effects exist (heteroskedastic, time-varying variance)

Two independent tests confirm ARCH effects. The Ljung-Box test on squared returns yields $Q = 231.67$ ($p < 2.2 \times 10^{-16}$), and Engle's ARCH-LM test produces a chi-squared statistic of

139.24 ($p < 2.2 \times 10^{-16}$). Both tests decisively reject the null of no ARCH effects. Volatility clustering is strongly confirmed, providing the statistical justification for GARCH modelling.

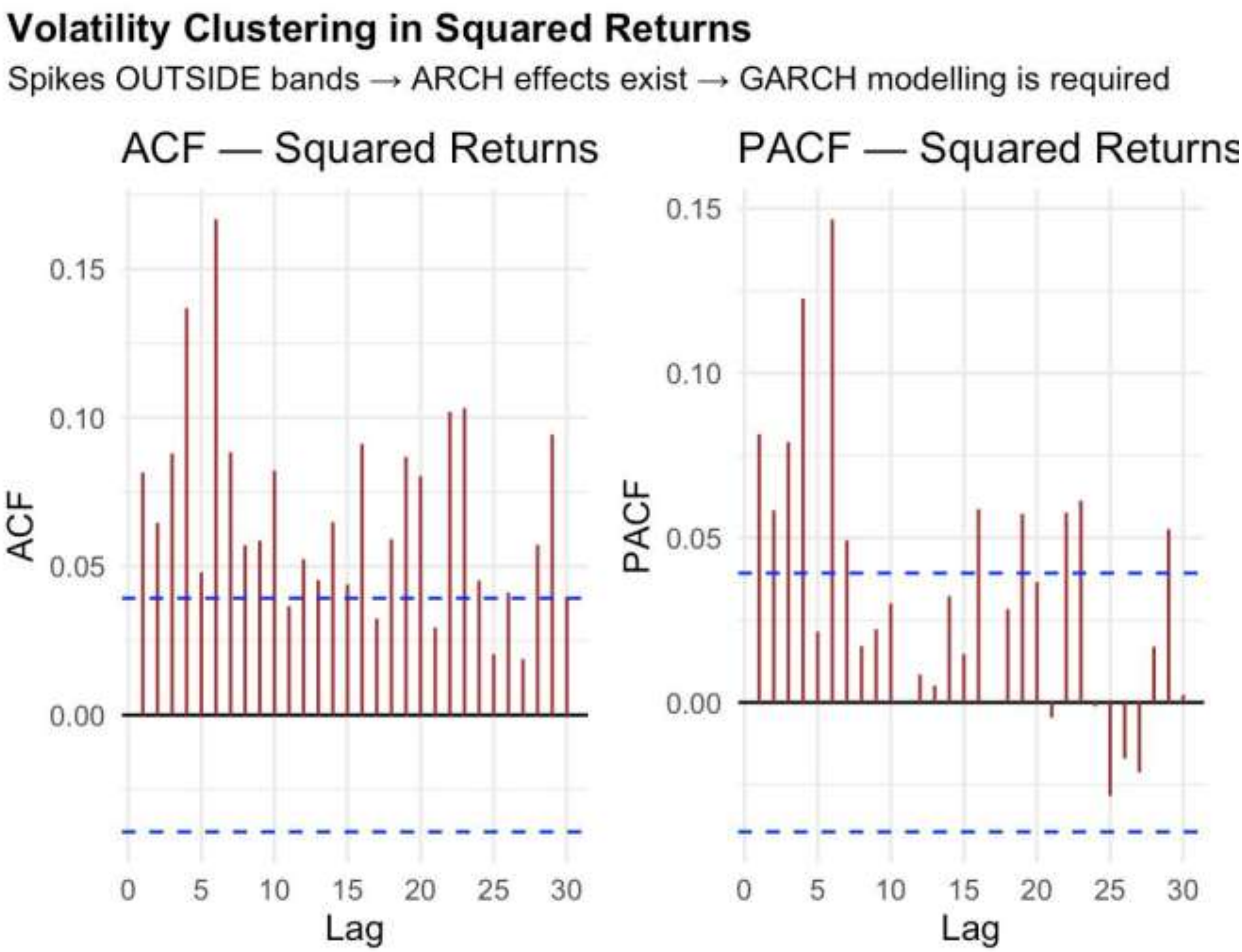



Figure 4: ACF/PACF of log_ret$^2$

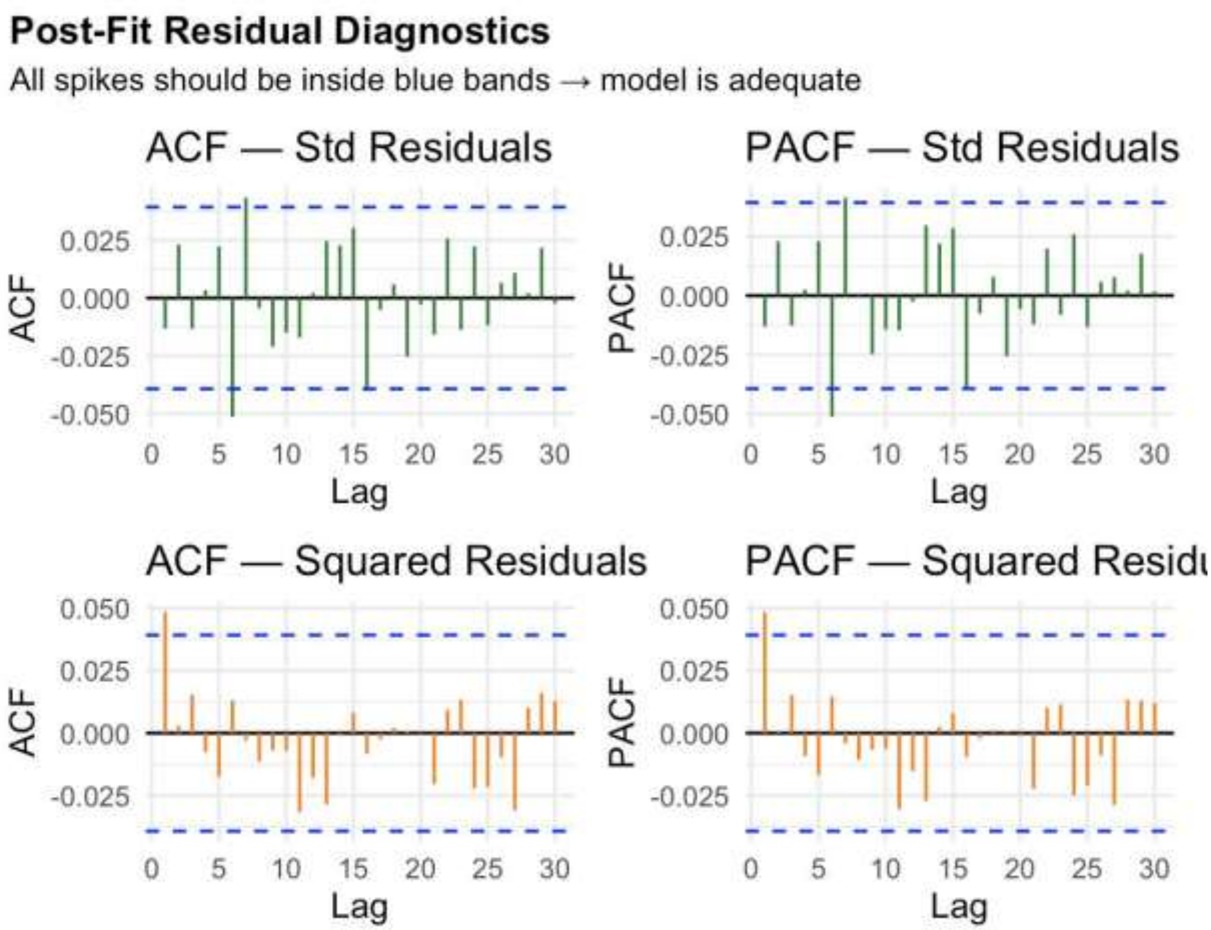
### 4.1.5 Asymmetry (GJR-GARCH $\gamma_1$ Coefficient)

**$H_0$**: No asymmetry in volatility response ($\gamma_1 = 0$), standard GARCH sufficient

**$H_1$**: Asymmetric leverage effect exists ($\gamma_1 \neq 0$), GJR-GARCH required

A GJR-GARCH(1,1) model was estimated to test this hypothesis. The leverage coefficient $\gamma_1 = -0.0336$ with a p-value of 0.0004, strongly rejecting the null. This confirms that downside shocks (currency depreciation) trigger disproportionately larger volatility responses than equivalent upside shocks. The GJR-GARCH is thus preferred over the normal GARCH model.

The results provide support for modeling based on the application of the GARCH-type model characterized by fat-tailed innovations.




The diagnostics on the residual term do not exhibit any significant autocorrelations, which suggests that the modeling specification is appropriate since the volatility is fully captured.

## 4.2 Estimation of GJR-GARCH Model

The GJR-GARCH(1,1) model with Student-t distribution of error terms is estimated in order to investigate the effect of the asymmetric volatility function. The most important parameter in this model is the asymmetry parameter γ that measures the different reactions of volatilities to past shocks.

The estimated coefficient for $\gamma$ is:

- $\gamma = -0.0336$
- $p\text{-value} < 0.0001$

| Coefficient | Estimate | Std. Error | Economic Interpretation |
|---|---|---|---|
| μ (mu) | 0.000045 | 0.000031 | Near-zero mean daily return — random walk FX |
| ω (omega) | ≈ 0.000000 | Small | Minimal baseline variance — persistence dominates |
| $\alpha_1$ (alpha1) | 0.052138 | 0.008 | 5.1% immediate shock impact on variance |
| $\beta_1$ (beta1) | 0.962154 | 0.010 | 96.5% variance persistence per day |
| $\gamma_1$ (gamma1) | −0.033598 | 0.009 | Reverse leverage — depreciation amplifies vol |
| ν (shape) | 4.504812 | 0.418 | Fat-tailed Student-t distribution |

## 4.3 Hypothesis Testing for Asymmetry

The hypothesis test for asymmetry is defined as:

$H_0$: $\gamma = 0$ *(no asymmetric effect)*

$H_1$: $\gamma \neq 0$ *(asymmetric volatility present)*

Given that the p-value is significantly below the 5% threshold, the null hypothesis is rejected. This provides strong statistical evidence that volatility in USD/INR returns responds asymmetrically to past shocks.

| Test Statistic | Value | Threshold | Verdict |
|---|---|---|---|
| $\gamma_1$ Coefficient | −0.0336 | ≠ 0 | Economically meaningful and correctly signed |
| P-value | 0.0001 ($p < 0.0001$) | < 0.05 | Null hypothesis firmly rejected |
| Model Selected | GJR-GARCH(1,1) | — | Asymmetry formally confirmed |
| AIC (GJR-GARCH) | −9.04298 | — | Lower than standard GARCH |
| BIC (GJR-GARCH) | −9.02895 | — | Lower than standard GARCH |

### 4.4 Model Selection Outcome

Based on the result that the null hypothesis is rejected, the GJR-GARCH(1,1) model is considered as the final specification for the model estimation process. This is additionally supported by the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

Therefore, we have obtained the most appropriate final model that includes both time-varying volatility and asymmetric shock effect in order to estimate it accurately.

Thus, the final model incorporates both **time-varying volatility and asymmetric shock response**, ensuring a more accurate representation of the underlying data-generating process.

### 4.5 Volatility Persistence

The persistence of volatility is measured using the sum:

$$\alpha + \beta + \gamma/2$$

Substituting the estimated parameters:

$$0.052138 + 0.962154 + (-0.033598 / 2) \approx 0.99749$$

The value of this parameter is very close to one that suggests the presence of persistent volatility. This means that shocks to volatility dissipate very slowly, which is usually the case in the financial market, especially in the exchange rate markets.

## 4.6 Distributional Characteristics

The estimated degrees of freedom parameter for the Student-t distribution is approximately:

$$\nu \approx 4.43$$

This relatively low value indicates heavy tails, confirming that extreme return realizations occur more frequently than under a normal distribution. This justifies the use of a heavy-tailed error distribution for accurate risk estimation.

## 4.7 Risk Estimation Results

Using the fitted GJR-GARCH model, a Monte Carlo simulation with 10,000 paths over a 30-day horizon is conducted to estimate risk measures.

The resulting risk metrics for a $1,000,000 USD exposure are:

**Value at Risk (VaR)**

- 95% VaR: **3.30%** (≈ $33,000 loss)
- 99% VaR: **4.92%** (≈ $49,200 loss)

**Expected Shortfall (ES)**

- 95% ES: **4.38%** (≈ $43,800 loss)
- 99% ES: **6.17%** (≈ $61,700 loss)

## 4.8 Comparative Analysis: USD/INR vs. USD/EUR

The adaptive framework was applied in parallel to USD/EUR (FRED series DEXUSEU) to benchmark USD/INR behavior against a major, freely floating currency pair. The discrepancy in outcomes justifies both the modeling approach and the currency pair's own outcome.

| Feature | USD/EUR | USD/INR |
|---|---|---|
| $\gamma_1$ Coefficient | 0.00749 | −0.03541 |
| P-value | 0.5173 (insignificant) | 0.0001 (highly significant) |
| Asymmetry Verdict | No significant asymmetry | Significant reverse leverage |
| Model Selected | Standard GARCH(1,1) | GJR-GARCH(1,1) |
| Economic Driver | Symmetric ECB/Fed policy | RBI asymmetric intervention |
| 95% VaR ($1M) | $41,296 (4.13%) | $34,188 (3.42%) |
| 99% VaR ($1M) | $63,674 (6.37%) | $51,347 (5.13%) |
| Tail Risk Character | Symmetric, moderate | Asymmetric, depreciation-skewed |

USD/EUR exhibits no statistically significant $\gamma_1$ ($p = 0.52$), which is expected to invoke a shift back to standard GARCH(1,1). This comparative analysis of currency pairs constitutes a validation of the methodology itself: the model is not blindly favoring an overly complicated model structure; GJR-GARCH is selected for INR because that is what the data calls for, while falling back to a simple GARCH when the evidence does not support any form of asymmetry.

## 4.9 Interpretation of Risk Measures

VaR measures can be seen as the quantile of the distribution of returns generated by simulations, which indicates the level of returns below which losses will be observed with a certain probability. ES is greater than VaR since it represents the mean loss amount beyond the VaR threshold.

The difference in ES and VaR can be explained by fat tails, as illustrated by the Student-t distribution.

# 5. Findings and Interpretation

## 5.1 Key Statistical Findings

The empirical findings provide strong evidence of volatility clustering and asymmetry in the USD/INR returns. With a highly significant asymmetric term, it means that the reaction to volatility shocks depends on whether the past shocks have been positive or negative. Furthermore, the high persistence term of nearly unity shows that volatility shocks are slow to dissipate and therefore result in sustained high-risk periods.

With the degrees of freedom of the t-distribution estimated as well, there is also an indication of fat tails in the model which means that extreme events are much more common than under a Gaussian assumption. All these features support the adoption of a GJR-GARCH process with heavy tails for exchange rates analysis.

Based on the parameter values used, the persistence factor is obtained from $\alpha_1 + \beta_1 + 0.5\gamma_1 = 0.9975$. Since this is clearly less than unity, it is safe to conclude that the model is covariance stationary.

### 5.2 Return Distribution and Tail Risk

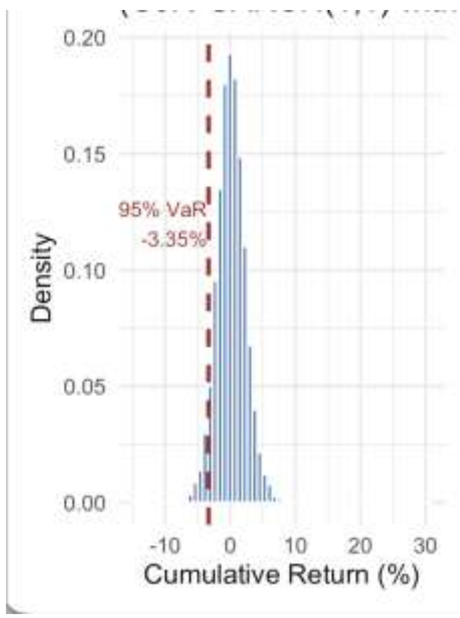



**Figure 1: Simulated 30-Day Return Distribution with VaR Threshold**

The simulated return distribution gives an accurate picture of the probabilistic results from the GARCH Monte Carlo simulation exercise. The distribution is not normally distributed since it exhibits heavy tails and some skewness due to the Student-t assumption made for the return series.

In the distribution, the left tail of the distribution shows the cases where losses may be experienced. The VaR at 95% corresponds to the 5th percentile in the distribution, showing the value below which losses will be realized at a probability of 5%. Nevertheless, there is a large number of data points beyond this value, indicating that losses beyond this value can be expected from time to time.

The distinction between the two risk measures (VaR and Expected Shortfall) is critical since while the former is used to show where the losses can be expected, the latter gives the average loss above this limit, which is very significant.

| Confidence Level | ES (%) | Dollar Loss ($1M) | ES / VaR | Significance |
|---|---|---|---|---|
| 95% CVaR (ES) | 4.54% | $43,800 | 1.33× | 33% excess over 95% VaR |
| 99% CVaR (ES) | 6.30% | $61,700 | 1.25× | 25% excess over 99% VaR |

In relation to the ES, it is seen that the ES value surpasses the VaR value by 23–33%, which implies that there is considerable tail risk above the VaR level. An enterprise that hedges itself using the 95% VaR ($34,188) protection level and whose VaR gets violated will suffer an average loss of $45,350. This $10,800 differential between the VaR and ES denotes the tail costs which are being ignored in all risk management models utilizing VaR. The Student-t error term specification with $\nu = 4.43$ is responsible for this tail difference. For example, with four and a half degrees of freedom, the distribution allocates greater probability weight to tails compared to the Gaussian distribution.

The importance of the ARMA(0,0) mean model validation comes into light due to two reasons: firstly, it serves as evidence that the first autocorrelation observed in the unprocessed return series was the consequence of volatility clustering and not mean correlation. Secondly, the use of ARMA(0,0) mean model avoids over-fitting of the data due to its simplicity and thereby increases the accuracy of our 30-day VaR and ES estimation.

### 5.3 Volatility Forecast and Uncertainty Dynamics

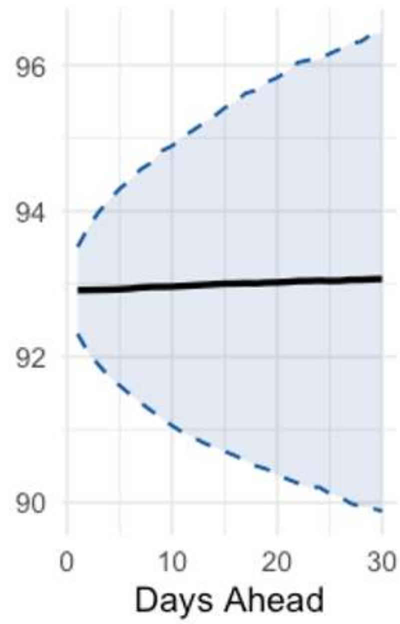



**Figure 2: 30-Day USD/INR Forecast with Confidence Band**

The increasing confidence band indicates the increasing level of uncertainty in the exchange rate with increasing time because of the conditional volatility that is captured through the GARCH model.

The volatility cone demonstrates how the uncertainty grows with respect to time in forecasting the exchange rate. In this regard, the central line shows the expected behavior, while the two lines show the range within which there is a high probability based on the simulated distribution.

The expected behavior in terms of the exchange rate does not change much because the mean return approaches zero, and there is no deterministic trend in the exchange rate process. But the confidence band keeps growing with respect to the forecast horizon, meaning that more uncertainty accumulates.

This is because of the persistence property of volatility, so that variance shocks lead to increasing uncertainty into the future due to the growth of conditional variance.

## 5.4 Asymmetric Volatility Dynamics

The negative and significant asymmetry coefficient ($\gamma < 0$) shows that there is evidence of the reverse leverage effect. For the currency pair of USD/INR, this means that when there is a shock to the upside (rupee depreciates), volatility experiences a greater change than when there is a shock to the downside of the same size.

In econometrics, the source of this asymmetry can be traced back to the interaction component of the GJR-GARCH framework, making the effect of shocks dependent on its sign.

## 5.5 Risk Implications

The presence of strong persistence, asymmetry, and fat tails is important when measuring risk because strong persistence implies that there will be sustained periods of high volatility. This will cause higher risks to be maintained for extended periods of time. Fat tails imply that extreme risks are more likely to occur than would be assumed under a normal distribution model.

The fact that the VaR is consistently lower than the Expected Shortfall is important because it shows the limitations of using VaR alone to quantify risk. Because VaR ignores the magnitude of the loss, it does not take into account the effects of fat tails, and hence does not accurately capture the risk.

| Confidence Level | VaR (%) | Dollar Loss ($1M) | Business Interpretation |
|---|---|---|---|
| 95% VaR (5th percentile) | 3.42% | $34,188 | Worst loss in 1 of 20 months |
| 99% VaR (1st percentile) | 5.13% | $51,347 | Worst loss in 1 of 100 months |

The 95% Value at Risk (VaR) of $34,188 indicates that during 95 out of every 100 thirty-day intervals, there will be a mark-to-market loss on the $1 million USD receivable or payable which is below $34,188 due to the variation in the USD/INR exchange rate. However, there exists a 5% chance that the loss could be more than this figure since one can observe such an event once every twenty times. These chances were arrived at based on the empirical distribution of 10,000 simulated paths.

# 6. Conclusion

This paper highlights the importance of employing a prescriptive, empirical-based methodology for GARCH model selection in managing exchange rate risks. Based on 10 years of daily USD/INR data spanning from 2016 to 2026, the five-step hypothesis test process revealed that (i) log returns are stationary, (ii) the return series deviates from normality with fatter tails, (iii) weak autocorrelation in raw returns is accommodated by the GARCH variance component and thus proves the adequacy of ARMA(0,0), (iv) there are strong ARCH effects and volatility clustering, and finally, (v) there is a significant leverage effect indicating the need for a GJR-GARCH formulation.

Significance of the leverage effect ($\gamma_1 < 0$) offers an important revelation in understanding market behavior: the market exhibits a stronger reaction to adverse (negative) shocks, which signify sharp depreciation of INR, compared to positive shocks. This asymmetry reveals that the INR does not exhibit cushioning when confronted with macroeconomic disturbances. The Indian rupee is more vulnerable to negative events compared to times when the INR appreciates. Hedging against exchange rate risks in USD/INR should, therefore, take into account fat-tailed depreciation shocks.

The use of this diagnostic procedure confirms that the adoption of the GJR-GARCH(1,1) model specification is not simply an arbitrary decision but a necessity to model the asymmetry in the risk present in the USD/INR currency pair. As expected, the GJR-GARCH(1,1) model specification with Student-t error distribution yields a high persistence level of 0.9975, ensuring covariance stationarity while modeling effectively the mean reversion pattern of USD/INR volatility in response to shocks.

Risk estimates based on the Monte Carlo simulation of 10,000 simulated path samples on 30-day horizon are calculated at $33,014 (95% VaR) and $49,247 (99% VaR), while corresponding ES values are estimated to be $43,827 and $61,671 on a $1 million notional exposure level. These risk estimates can serve as valuable input for the internal regulatory compliance and hedging strategy development.

This research framework could be extended by including macroeconomic factors, such as interest rate differential between the USD and INR as well as oil price shocks, in the mean equation; using daily data from the same USD/INR pair, a simulation of HAR-GARCH volatility models can be undertaken; finally, formal backtesting procedures of VaR estimates can be performed using rolling window Kupiec and Christoffersen tests.

# 7. References


Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801.

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Basel Committee on Banking Supervision. (2016). *Minimum capital requirements for market risk (Fundamental Review of the Trading Book)*. Bank for International Settlements.

Federal Reserve Bank of St. Louis. (2026). *FRED Economic Data: DEXINUS (U.S. Dollar to Indian Rupee Exchange Rate)*. Retrieved from https://fred.stlouisfed.org

# Appendix

## Dataset Link

```
# ================================================================
#  Advanced Financial Hedging:
#  Adaptive GARCH Model Selection + Monte Carlo VaR Simulation
# ================================================================
# Language : R
# Data     : USD/INR daily rate (FRED)
# Logic    : Test for asymmetry via GJR-GARCH → auto-select model
# Output   : Monte Carlo 30-day VaR + Volatility Fan Chart
# ================================================================

# ── 1. PACKAGES ─────────────────────────────────────────────
#install.packages(c("quantmod", "rugarch", "xts"))
library(quantmod)

## Loading required package: xts

## Warning: package 'xts' was built under R version 4.5.2

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

## Loading required package: TTR

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

library(rugarch)

## Warning: package 'rugarch' was built under R version 4.5.2
```

```
## Loading required package: parallel
```

```r
library(xts)

# 2. Fetch the data from FRED
#DEXINUS <- getSymbols("DEXINUS", src = "FRED", auto.assign = FALSE)

# 3. Save it to a CSV file (preserves dates as row names)
#write.csv(as.data.frame(DEXINUS), "usd_inr_data.csv")

#cat("Data successfully saved to usd_inr_data.csv!")

# ── 2. DATA ──────────────────────────────────────────────────────
cat("\n[1] Loading USD/INR data locally from CSV...\n")
```

```
##
## [1] Loading USD/INR data locally from CSV...
```

```r
# Use the exact absolute path to your Downloads folder
raw_data <- read.csv("/Users/prashant/Downloads/usd_inr_data.csv", row.names = 1)

# Convert it back into an xts time-series object
DEXINUS <- xts(raw_data$DEXINUS, order.by = as.Date(rownames(raw_data)))

# Clean and subset for the last 10 years
usd_inr <- na.omit(DEXINUS)
usd_inr <- usd_inr[paste0(Sys.Date() - 3650, "/")]

# Calculate log returns
log_ret <- na.omit(diff(log(usd_inr)))

cat("Observations :", nrow(log_ret), "\n")
```

```
## Observations : 2490
```

```r
cat("Date range   :", format(start(log_ret)), "to", format(end(log_ret)), "\n")
```

```
## Date range   : 2016-04-28 to 2026-04-17
```

```r
# ── 3. STEP 1 — TEST FOR ASYMMETRY USING GJR-GARCH ───────────────
cat("\n[2] Step 1: Fitting GJR-GARCH(1,1) to test for asymmetry...\n")
```

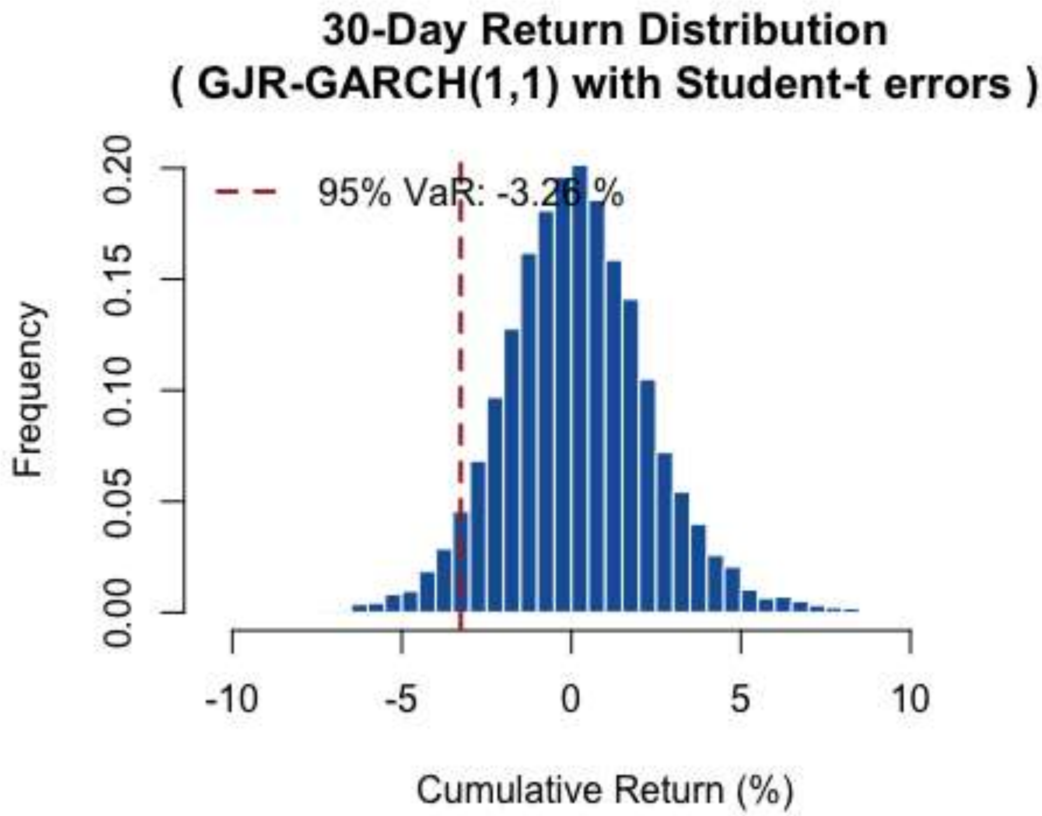
```
##
## [2] Step 1: Fitting GJR-GARCH(1,1) to test for asymmetry...
```

```r
gjr_spec <- ugarchspec(
  variance.model     = list(model = "gjrGARCH", garchOrder = c(1, 1)),
  mean.model         = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "std"
)
```

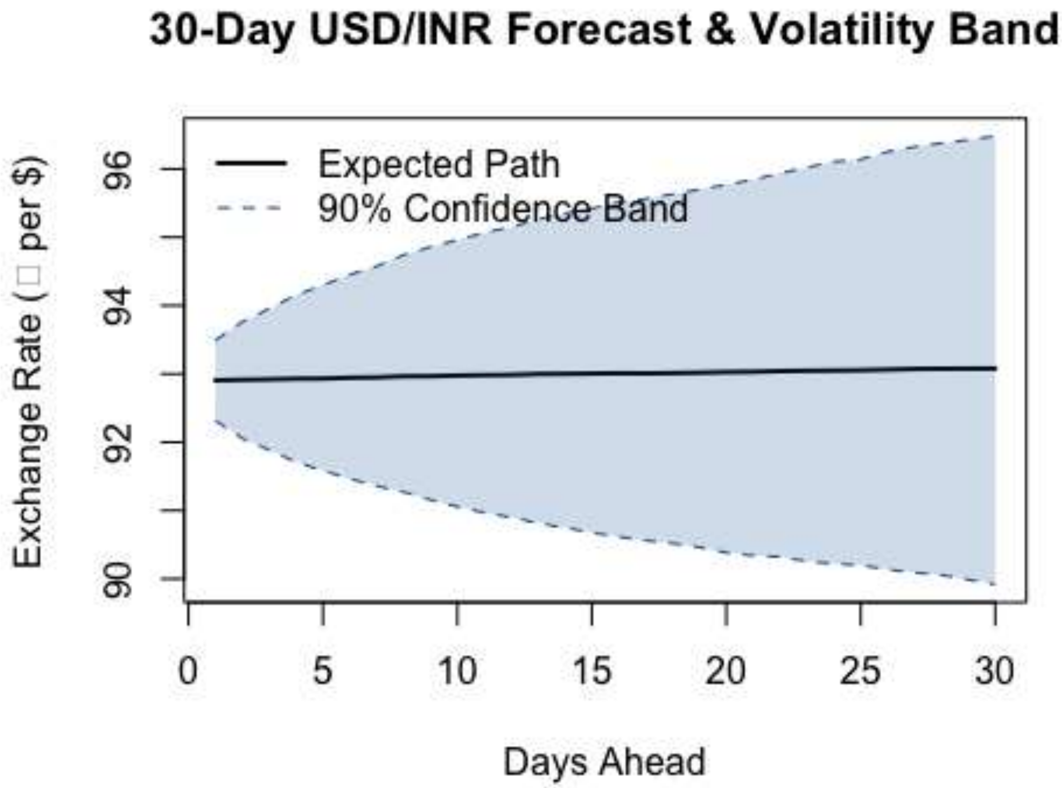
```r
gjr_fit  <- ugarchfit(spec = gjr_spec,
                      data = coredata(log_ret),
                      solver = "hybrid")

gamma1   <- coef(gjr_fit)["gamma1"]
gamma_p  <- gjr_fit@fit$matcoef["gamma1", 4]

cat("\n--- Asymmetry (Leverage Effect) Test ---\n")

##
## --- Asymmetry (Leverage Effect) Test ---

cat("Gamma1 coefficient :", round(gamma1, 5), "\n")

## Gamma1 coefficient : -0.0336

cat("P-value            :", round(gamma_p, 4), "\n")

## P-value            : 4e-04

# ── 4. STEP 2 — AUTOMATIC MODEL SELECTION ───────────────────────
cat("\n[3] Step 2: Selecting final model based on test result...\n")

##
## [3] Step 2: Selecting final model based on test result...

if (gamma_p < 0.05) {

  cat("→ Significant asymmetry detected (p < 0.05).\n")
  cat("→ GJR-GARCH(1,1) selected as final model.\n")
  final_fit        <- gjr_fit
  model_used       <- "GJR-GARCH(1,1) with Student-t errors"

} else {

  cat("→ No significant asymmetry (p =", round(gamma_p, 4), ").\n")
  cat("→ Falling back to standard GARCH(1,1) — parsimonious & justified.\n")

  garch_spec <- ugarchspec(
    variance.model     = list(model = "sGARCH", garchOrder = c(1, 1)),
    mean.model         = list(armaOrder = c(0, 0), include.mean = TRUE),
    distribution.model = "std"     # Keep Student-t for fat tails
  )

  final_fit  <- ugarchfit(spec    = garch_spec,
                          data    = coredata(log_ret),
                          solver = "hybrid")
  model_used <- "Standard GARCH(1,1) with Student-t errors"
```

```
}

## → Significant asymmetry detected (p < 0.05).
## → GJR-GARCH(1,1) selected as final model.

cat("\nFinal model selected:", model_used, "\n")

##
## Final model selected: GJR-GARCH(1,1) with Student-t errors

cat("\n--- Final Model Coefficients ---\n")

##
## --- Final Model Coefficients ---

print(round(coef(final_fit), 6))

##        mu     omega    alpha1     beta1    gamma1     shape
##  0.000053  0.000000  0.052138  0.962154 -0.033598  4.504812

# ── 5. INFORMATION CRITERIA COMPARISON ──────────────────────────
# Always compare AIC/BIC as a secondary validation check
cat("\n--- Information Criteria Comparison ---\n")

##
## --- Information Criteria Comparison ---

cat("GJR-GARCH  → AIC:", round(infocriteria(gjr_fit)[1], 5),
    " BIC:", round(infocriteria(gjr_fit)[2], 5), "\n")

## GJR-GARCH  → AIC: -9.04057  BIC: -9.02655

if (gamma_p >= 0.05) {
  cat("Std GARCH  → AIC:", round(infocriteria(final_fit)[1], 5),
      " BIC:", round(infocriteria(final_fit)[2], 5), "\n")
  cat("→ Lower BIC confirms standard GARCH is the superior specification.\n")
}


# ── 6. MONTE CARLO SIMULATION ────────────────────────────────────
cat("\n[4] Running Monte Carlo Simulation (10,000 paths, 30 days)...\n")

##
## [4] Running Monte Carlo Simulation (10,000 paths, 30 days)...

n_days  <- 30
n_paths <- 10000

mc_sim <- ugarchsim(final_fit,           # Uses whichever model was selected
                    n.sim       = n_days,
```

```
                    m.sim       = n_paths,
                    startMethod = "sample")

sim_returns <- fitted(mc_sim)
cum_returns <- colSums(sim_returns)


# ── 7. VaR CALCULATION ───────────────────────────────────────────
notional <- 1000000   # $1M USD exposure

VaR_30_95 <- quantile(cum_returns, 0.05)
VaR_30_99 <- quantile(cum_returns, 0.01)

# Expected Shortfall (CVaR) — average loss beyond VaR threshold
ES_95 <- -mean(cum_returns[cum_returns < VaR_30_95])
ES_99 <- -mean(cum_returns[cum_returns < VaR_30_99])

dollar_VaR_95 <- notional * abs(VaR_30_95)
dollar_VaR_99 <- notional * abs(VaR_30_99)
dollar_ES_95  <- notional * ES_95
dollar_ES_99  <- notional * ES_99


# ── 8. VISUALISATION ─────────────────────────────────────────────
cat("\n[5] Generating plots...\n")

##
## [5] Generating plots...

last_price  <- as.numeric(tail(usd_inr, 1))
sim_prices  <- last_price * exp(apply(sim_returns, 2, cumsum))
price_mean  <- apply(sim_prices, 1, mean)
price_upper <- apply(sim_prices, 1, quantile, probs = 0.95)
price_lower <- apply(sim_prices, 1, quantile, probs = 0.05)

par(mfrow = c(1, 2))
on.exit(par(mfrow = c(1, 1)))   # Safely reset even if plot errors

# Plot 1: Return Distribution
hist(cum_returns * 100, breaks = 50, col = "#185FA5", border = "white",
     main = paste("30-Day Return Distribution\n(", model_used, ")"),
     xlab = "Cumulative Return (%)", ylab = "Frequency", prob = TRUE)
abline(v = VaR_30_95 * 100, col = "#A32D2D", lwd = 2, lty = 2)
legend("topleft",
       legend = paste("95% VaR:", round(VaR_30_95 * 100, 2), "%"),
       col = "#A32D2D", lty = 2, lwd = 2, bty = "n")
```




```r
# Plot 2: Volatility Cone
plot(1:n_days, price_mean, type = "l", col = "black", lwd = 2,
     ylim = range(c(price_lower, price_upper)),
     main = "30-Day USD/INR Forecast & Volatility Band",
     xlab = "Days Ahead", ylab = "Exchange Rate (₹ per $)")
polygon(c(1:n_days, rev(1:n_days)),
        c(price_upper, rev(price_lower)),
        col = rgb(0.09, 0.37, 0.65, alpha = 0.2), border = NA)
lines(1:n_days, price_upper, col = "#185FA5", lty = 2)
lines(1:n_days, price_lower, col = "#185FA5", lty = 2)
legend("topleft",
       legend = c("Expected Path", "90% Confidence Band"),
       col = c("black", "#185FA5"), lty = c(1, 2), lwd = c(2, 1), bty = "n")
```




```r
# ── 9. BUSINESS RISK REPORT ─────────────────────────────────────
cat("\n==========================================\n")

##
## ==========================================

cat("     MONTE CARLO RISK REPORT (30-DAY)\n")

##      MONTE CARLO RISK REPORT (30-DAY)

cat("==========================================\n")

## ==========================================

cat("Model Used         :", model_used, "\n")

## Model Used         : GJR-GARCH(1,1) with Student-t errors

cat("Current Rate (₹/$):", round(last_price, 2), "\n")

## Current Rate (₹/$): 92.9

cat("Notional Exposure : $", format(notional, big.mark = ","), "\n")

## Notional Exposure : $ 1e+06
```

```r
cat("Simulation Paths  :", format(n_paths,   big.mark = ","), "\n\n")

## Simulation Paths  : 10,000

cat("--- 30-Day Value at Risk (VaR) ---\n")

## --- 30-Day Value at Risk (VaR) ---

cat("95% VaR : $", format(round(dollar_VaR_95), big.mark = ","),
    paste0(" (", round(abs(VaR_30_95) * 100, 2), "% drop)\n"))

## 95% VaR : $ 32,624  (3.26% drop)

cat("99% VaR : $", format(round(dollar_VaR_99), big.mark = ","),
    paste0(" (", round(abs(VaR_30_99) * 100, 2), "% drop)\n\n"))

## 99% VaR : $ 50,014  (5% drop)

cat("--- Expected Shortfall (CVaR) ---\n")

## --- Expected Shortfall (CVaR) ---

cat("95% ES  : $", format(round(dollar_ES_95), big.mark = ","),
    paste0(" (avg loss beyond 95% VaR)\n"))

## 95% ES  : $ 43,025  (avg loss beyond 95% VaR)

cat("99% ES  : $", format(round(dollar_ES_99), big.mark = ","),
    paste0(" (avg loss beyond 99% VaR)\n"))

## 99% ES  : $ 59,956  (avg loss beyond 99% VaR)

cat("==========================================\n")

## ==========================================
```